Module 5 slides: Oral Absorption (worked examples)

Module 5 - Single Oral Administration · Lecture slides, worked examples · 16 Sep 2026

Name _____ Date _____

Example 1 An investigational new drug has a half-life of absorption of 45 minutes. What is its absorption rate constant?

Example 1 The same investigational drug has a half-life of elimination of 3 hours. What is its elimination rate constant?

Example 1 The following pharmacokinetic parameters were found for an investigational new drug: V_D is 22 L, the half-life of absorption is 45 minutes and the half-life of elimination is 3 hours. The drug is 85% bioavailable and follows a one-compartment model. When does the maximum concentration occur after a single 500-mg oral dose?

Example 1 For the same investigational drug, with V_D of 22 L, an absorption half-life of 45 minutes, an elimination half-life of 3 hours and 85% bioavailability, what maximum concentration is expected after a single 500-mg oral dose?

Example 1 A 500-mg oral dose of an investigational drug with V_D of 22 L, an absorption half-life of 45 minutes, an elimination half-life of 3 hours and 85% bioavailability produces a peak concentration of 12.17 mg/L. What peak concentration is expected if the dose is increased to 1000 mg?

Example 2 A single oral dose of an antibiotic was given to an adult 70 kg male patient. The drug fits a one-compartment open model with $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mcg/mL and t in hours. What is the elimination half-life for this drug in this patient?

Example 2 For the same antibiotic, described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$ with t in hours, what is the half-life of absorption?

Example 2 A single oral dose of an antibiotic given to an adult 70 kg male patient is described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mcg/mL and t in hours. Calculate t_{max}.

Example 2 A single oral dose of an antibiotic given to an adult 70 kg male patient is described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mcg/mL and t in hours. Calculate C_{max}.

Example 2 The antibiotic described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with concentration in mg/L and time in hours, was given as a 500-mg oral dose that is 87% bioavailable. What is the apparent volume of distribution?

Example 2 The antibiotic described by $C_p = 75(e^{-0.22t} - e^{-2.75t})$, with C_p in mg/L and t in hours, was given as a single oral dose. What plasma concentration is expected 4 hours after the dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Example 1 Her key

Answer: 0.924 hr^{-1}

WHAT THE STEM GIVES

t½a 45 minutes becomes 0.75 hr, the denominator of $\frac{0.693}{t½a}$

THE WORKING, ONE LINE AT A TIME

UNIT

$$45 \text{ min} \times \frac{1 \text{ hr}}{60 \text{ min}} = 0.75 \text{ hr}$$

Every other parameter in this problem is quoted in hours, and a rate constant taken from a half-life in minutes would come out in reciprocal minutes and could not be combined with them. Minutes multiplied by hours per minute cancels the minutes and leaves hours. Forty-five minutes is three quarters of an hour.

SETUP

$$t½a = \frac{0.693}{k_a}, \text{ so } k_a = \frac{0.693}{t½a}$$

Absorption is a first-order process, so its half-life and its rate constant are linked by 0.693 in exactly the same way as on the elimination side. The half-life given is an absorption half-life, so the constant that comes out is k_a and not k.

ALGEBRA

$$k_a = \frac{0.693}{0.75 \text{ hr}} = 0.924 \text{ hr}^{-1}$$

Dividing a pure number by a time in hours leaves reciprocal hours, which is the unit every first-order rate constant must carry. The arithmetic is 0.693 divided by 0.75.

ROUND

$$k_a = 0.924 \text{ hr}^{-1}$$

Her worked value is 0.924 per hour, carried to three decimal places, which is the precision she uses for rate constants throughout this example.

> **DOES THE NUMBER MAKE SENSE?**
> 45 minutes is 0.75 hr, so k_a is 0.693 over 0.75, a little under 1 per hour: 0.924 hr^{-1}. A half-life under an hour means k_a above 0.693.

Example 1 Her key

Answer: 0.231 hr^{-1}

WHAT THE STEM GIVES

t½ 3 hours the denominator of $\frac{0.693}{t½}$

THE WORKING, ONE LINE AT A TIME

SETUP

$t_{½} = \frac{0.693}{k}$, so $k = \frac{0.693}{t_{½}}$

Elimination is first order, so the half-life is 0.693 divided by the rate constant, and this relation is not on the equation sheet and has to be held from memory. Rearranging it isolates the rate constant.

ALGEBRA

$k = \frac{0.693}{3\ \text{hr}} = 0.231\ \text{hr}^{-1}$

The half-life is already in hours, so no conversion is needed and the division can be done directly. A pure number over hours leaves reciprocal hours.

ROUND

$k = 0.231\ \text{hr}^{-1}$

Her worked value is 0.231 per hour, to three decimal places, matching the precision she uses for the absorption constant in the same problem.

> **DOES THE NUMBER MAKE SENSE?**
> k is 0.693 over 3, 0.231 hr^{-1}; a half-life over an hour means k below 0.693, and 3 × 0.231 returns 0.693.

Example 1 Her key

Answer: 2 hr

WHAT THE STEM GIVES

V_D	22 L	*not needed: t_{max} depends on k_a and k only*
$t_{½a}$	45 minutes	becomes 0.75 hr, so $k_a = \frac{0.693}{0.75} = 0.924\ \text{hr}^{-1}$
$t_{½}$	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$
F	85%	*not needed: it scales the concentration, not the time*
D_0	500-mg	*not needed: the dose scales the concentration, not the time*

THE WORKING, ONE LINE AT A TIME

UNIT

45 min = 0.75 hr; the elimination half-life is already 3 hr

The two half-lives are given in different units and the expression for the time to peak subtracts one rate constant from another, which is only meaningful if both are in the same reciprocal time. Converting the 45 minutes to 0.75 hour before anything else is done keeps every later line in hours.

SETUP

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

The peak is the moment the rising and falling influences balance, and setting the derivative of the concentration equation to zero and solving for time gives this expression. Only the two rate constants survive, because the dose, the bioavailability and the volume of distribution multiply the whole curve and cancel.

ALGEBRA

$$k_a = \frac{0.693}{0.75\ \text{hr}} = 0.924\ \text{hr}^{-1};\ k = \frac{0.693}{3\ \text{hr}} = 0.231\ \text{hr}^{-1}$$

The time to peak is written in terms of the rate constants, not the half-lives, so both stated half-lives have to be converted. Each conversion is 0.693 divided by that process's own half-life.

ALGEBRA

$$t_{max} = \frac{\ln(0.924 / 0.231)}{(0.924 - 0.231)\ \text{hr}^{-1}} = \frac{\ln(4)}{0.693\ \text{hr}^{-1}}$$

The ratio of the two rate constants is a pure number, here exactly four, so its logarithm is dimensionless. Dividing that by a quantity in reciprocal hours leaves hours, which is what a time must be.

ALGEBRA

$$t_{max} = \frac{1.3863}{0.693\ \text{hr}^{-1}} = 2.0004\ \text{hr}$$

A dimensionless logarithm divided by a quantity in reciprocal hours leaves hours, which is what a time to peak must be. The ratio of the two rate constants is exactly 4 here, so the numerator is the natural logarithm of 4 and the arithmetic comes out very close to a whole number.

ROUND

$t_{max} = 2$ hours

Her worked value is 2 hours, and the unrounded 2.0004 rounds to it. The time to peak depends only on the two rate constants, so the 22 L volume, the 500 mg dose and the 85 per cent bioavailability given in the stem play no part in this particular answer.

> **DOES THE NUMBER MAKE SENSE?**
> k_a is four times k, so t_{max} is ln(4) over ($k_a - k$), 1.3863 over 0.693, 2 hr; the peak must come after the 45-minute absorption half-life.

Example 1 Her key

Answer: 12.17 mg/L

WHAT THE STEM GIVES

V_D	22 L	in the denominator of the prefactor: 22 × 0.693 = 15.246
t½a	45 minutes	becomes 0.75 hr, so $k_a = \frac{0.693}{0.75} = 0.924\ \text{hr}^{-1}$
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$; $k_a - k = 0.693$
F	85%	as 0.85 in the numerator of the prefactor

D_0	500-mg	in the numerator: 0.85 × 500 × 0.924 = 392.7

THE WORKING, ONE LINE AT A TIME

UNIT

$$45 \text{ min} = 0.75 \text{ hr, so } k_a = \frac{0.693}{0.75 \text{ hr}} = 0.924 \text{ hr}^{-1}; \; k = \frac{0.693}{3 \text{ hr}} = 0.231 \text{ hr}^{-1}$$

Both rate constants must be in reciprocal hours before they can be subtracted from one another or multiplied by a time in hours. The absorption half-life is the one given in minutes, so it is the one that needs converting.

SETUP

$$C_p = \frac{F \cdot k_a \cdot D_0}{V_D(k_a - k)}(e^{-k \cdot t} - e^{-k_a \cdot t}), \text{ evaluated at } t = t_{max}$$

The maximum concentration is simply the concentration at the turning point, so the full single oral dose equation is used with the time to peak in both exponentials. F scales the dose down to the amount that actually reaches the circulation and the volume of distribution converts that amount into a concentration.

ALGEBRA

$$t_{max} = \frac{\ln(0.924 / 0.231)}{(0.924 - 0.231) \text{ hr}^{-1}} = 2.0004 \text{ hr}$$

There is no separate expression for the peak concentration, so the time at which it occurs must be found first and then substituted. This is the step that is skipped when the peak is asked for without the time being asked for as well.

ALGEBRA

$$\text{Prefactor} = \frac{(0.85)(500 \text{ mg})(0.924 \text{ hr}^{-1})}{[(22 \text{ L})(0.693 \text{ hr}^{-1})]} = \frac{392.7}{15.246} = 25.758 \text{ mg/L}$$

Milligrams multiplied by reciprocal hours, divided by litres multiplied by reciprocal hours, cancels the reciprocal hours and leaves milligrams per litre. The 0.693 in the denominator here is the difference between the two rate constants, which coincides numerically with the constant used for half-lives and must not be confused with it.

ALGEBRA

$$\text{Bracket} = e^{-(0.231)(2)} - e^{-(0.924)(2)} = 0.62999 - 0.15751 = 0.47248$$

Each exponent is a reciprocal time multiplied by a time and is therefore dimensionless, as an exponent must be. The bracket is a pure fraction less than one, which is why the peak concentration is always smaller than the prefactor.

ALGEBRA

$$C_{max} = (25.758 \text{ mg/L})(0.47248) = 12.1701 \text{ mg/L}$$

Multiplying a concentration by a dimensionless fraction leaves a concentration. The bracket is well below one because the two exponentials partly cancel, which is why the peak is far lower than the prefactor on its own would suggest.

ROUND

$$C_{max} = 12.17 \text{ mg/L}$$

Her worked value is 12.17 milligrams per litre, to two decimal places, obtained by entering the whole expression in one pass. Entering it stepwise with the rounded intermediates shown here gives the same figure, so the order of the keystrokes does not matter.

DOES THE NUMBER MAKE SENSE?

The prefactor is 25.758 mg/L and the bracket at the peak is 0.47248, under half, so C_{max} is a little under half of 25.758, about 12.17 mg/L; it must be below the level the whole absorbed dose would give in 22 L.

Example 1 Worked by the drill

Answer: 24.34 mg/L

WHAT THE STEM GIVES

D_0	500-mg	the dose that gave the known peak: the denominator of the ratio
V_D	22 L	*not needed: unchanged, it cancels in the ratio*
t½a	45 minutes	*not needed: unchanged, so t_{max} and the bracket are unchanged*
t½	3 hours	*not needed: unchanged*
F	85%	*not needed: unchanged, it cancels in the ratio*
C_{max}	12.17 mg/L	the known peak, multiplied by the dose ratio
D_0 new	1000 mg	the numerator of the ratio $\frac{1000}{500} = 2$

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{max} = \frac{F \cdot k_a \cdot D_0}{V_D(k_a - k)}(e^{-k \cdot t} - e^{-k_a \cdot t})$ at $t = t_{max}$, in which the dose appears only in the prefactor, so C_{max} scales directly with D_0

Everything that fixes the shape of the curve, the two rate constants and therefore the time to peak, is untouched by the size of the dose, and the dose enters as a simple multiplier in front. That is what makes this a proportion rather than a fresh calculation.

ALGEBRA

$t_{max} = \frac{\ln(0.924 / 0.231)}{(0.924 - 0.231)\ hr^{-1}} = 2.0$ hr, unchanged by the dose

The dose does not appear in the expression for the time to peak, so doubling it cannot move the peak in time. The larger dose peaks at the same moment as the smaller one, directly above it on the curve.

ALGEBRA

The dose appears only in the prefactor, so C_{max} is proportional to D_0

F, k_a, V_D and the difference of the rate constants are all unchanged, and the bracket depends only on the rate constants and the time, which are also unchanged. The dose is therefore the only quantity that has moved, and it multiplies the whole expression.

ALGEBRA

$$C_{max} = (12.17 \text{ mg/L}) \times \frac{1000 \text{ mg}}{500 \text{ mg}} = (12.17 \text{ mg/L})(2)$$

The ratio of the doses is a pure number, so multiplying the old peak by it gives the new peak in the same units. This is quicker and less error-prone than re-evaluating the whole equation.

ROUND

$$C_{max} = 24.34 \text{ mg/L}$$

Recomputing the full equation with a 1000 mg dose gives 24.339 mg/L, which confirms the proportional route. Two decimal places matches the precision of the 12.17 mg/L she reports for the original dose.

> **DOES THE NUMBER MAKE SENSE?**
> The dose sits only in the prefactor, so doubling 500 mg to 1000 mg doubles the peak: 2 × 12.17 = 24.34 mg/L, at the same t_{max} of 2 hr.

Example 2 Her key

Answer: 3.15 hr

WHAT THE STEM GIVES

weight	70 kg	*not needed*
coefficient	75	*not needed: the half-life comes from an exponent*
k	0.22	the smaller exponent, in hr^{-1}, the denominator of $\frac{0.693}{k}$
k_a	2.75	*not needed: the larger exponent is absorption*

THE WORKING, ONE LINE AT A TIME

SETUP

The smaller rate constant is k, so $k = 0.22 \text{ hr}^{-1}$

For an ordinary oral drug absorption is much the faster process, so of the two constants in the exponents the larger belongs to absorption and the smaller to elimination. An unqualified half-life means the elimination half-life, so the constant needed here is 0.22 per hour and not 2.75 per hour.

ALGEBRA

$$t½ = \frac{0.693}{0.22 \text{ hr}^{-1}}$$

Elimination is first order, so its half-life is 0.693 divided by its rate constant. Dividing a pure number by reciprocal hours leaves hours.

ROUND

$$t½ = 3.15 \text{ hr}$$

The division gives exactly 3.15 hours at the two decimal places she reports half-lives to in this module. The 70 kg weight in the stem plays no part in this calculation.

> **DOES THE NUMBER MAKE SENSE?**

> The smaller exponent, 0.22 hr^{-1}, is k; 0.693 over 0.22 is a little over 3 hr, so 3.15 hr fits, and it must be longer than the absorption half-life from 2.75.

Example 2 Her key

Answer: 0.252 hr

WHAT THE STEM GIVES

coefficient	75	*not needed: the half-life comes from an exponent*
k	0.22	*not needed: the smaller exponent is elimination*
k_a	2.75	the larger exponent, in hr^{-1}, the denominator of $\frac{0.693}{k_a}$

THE WORKING, ONE LINE AT A TIME

SETUP

The larger rate constant is k_a, so $k_a = 2.75\ hr^{-1}$

Absorption is the faster of the two processes for an ordinary oral product, so the larger of the two exponential constants is the absorption rate constant. The question names absorption explicitly, so this is the constant to use rather than the one that would be used for an unqualified half-life.

ALGEBRA

$$t½a = \frac{0.693}{2.75\ hr^{-1}}$$

Absorption is first order, so its half-life is 0.693 divided by its own rate constant, exactly as on the elimination side. Dividing a pure number by reciprocal hours leaves hours.

ROUND

$t½a = 0.252$ hr

0.693 divided by 2.75 hr^{-1} gives 0.252 hour, which is about 15 minutes. An absorption half-life this short means absorption is nearly complete within the first hour: 94 per cent of the dose is absorbed by then.

> **DOES THE NUMBER MAKE SENSE?**
> The larger exponent, 2.75 hr^{-1}, is k_a; 0.693 over 2.75 is about a quarter hour, 0.252 hr, about 15 minutes, far shorter than the elimination half-life.

Example 2 Her key

Answer: 1 hr

WHAT THE STEM GIVES

weight	70 kg	*not needed*
coefficient	75	*not needed: t_{max} depends on k_a and k only*
k	0.22	the smaller exponent, in hr^{-1}: the denominator of the ratio, subtracted below
k_a	2.75	the larger exponent, in hr^{-1}: the numerator of the ratio; $k_a - k = 2.53$

THE WORKING, ONE LINE AT A TIME

SETUP

$k_a = 2.75\ hr^{-1}$ and $k = 0.22\ hr^{-1}$, read from the exponents by size

The two rate constants are already in the equation and need no conversion; only their identification is required. The larger is absorption because absorption is the faster process for an ordinary oral product, and putting them the other way round would give a negative denominator and a negative time.

SETUP

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

The time to peak is the moment at which the absorption and elimination rates are equal, obtained by setting the derivative of the concentration equation to zero. Only the rate constants appear, so the prefactor of 75 plays no part in this step.

ALGEBRA

$$t_{max} = \frac{\ln(2.75 / 0.22)}{(2.75 - 0.22)\ hr^{-1}} = \frac{\ln(12.5)}{2.53\ hr^{-1}}$$

The ratio of the two constants is dimensionless, so the logarithm can be taken. Dividing by a difference in reciprocal hours returns hours.

ALGEBRA

$$t_{max} = \frac{2.5257}{2.53\ hr^{-1}} = 0.9983\ hr$$

The ratio of the two rate constants is 12.5 and its natural logarithm is 2.5257, which is then divided by their difference of 2.53 reciprocal hours. That the numerator and denominator are so nearly equal here is a coincidence of these particular constants, not a general result.

ROUND

$t_{max} = 1$ hour

Her worked value is one hour, and the unrounded figure of 0.9983 hour rounds to it. The peak therefore occurs about one hour after the dose, which is early because absorption is more than twelve times faster than elimination for this drug.

> **DOES THE NUMBER MAKE SENSE?**
> k_a over k is 12.5, so ln(12.5) = 2.5257 over 2.53 is just under 1 hr; the peak must come after the quarter-hour absorption half-life and before much elimination at $k = 0.22\ hr^{-1}$.

Example 2 Worked by the drill

Answer: 55.4 mg/L

WHAT THE STEM GIVES

weight	70 kg	*not needed*
coefficient	75	in mg/L, multiplied by the bracket at t_{max}
k	0.22	the smaller exponent, in hr^{-1}: gives t_{max} and the $e^{-0.22}$ term
k_a	2.75	the larger exponent, in hr^{-1}: gives t_{max} and the $e^{-2.75}$ term

THE WORKING, ONE LINE AT A TIME

SETUP

C_{max} is the concentration at t_{max}, so the time to peak has to be found first and then substituted back into the equation the stem supplies

The peak is not a separate parameter to look up; it is the concentration equation evaluated at the one time the curve turns over. So the problem runs in two stages, and the second cannot start until the first gives a number. Naming both stages first stops the two rate constants being swapped.

ALGEBRA

$$t_{max} = \frac{\ln(2.75 / 0.22)}{(2.75 - 0.22)\ hr^{-1}} = 0.9983 \text{ hr, taken as 1 hr}$$

The peak concentration is the concentration at the time of the peak, so that time has to be in hand before the equation can be evaluated. This step is required even when the question asks only for the peak.

SETUP

$$C_{max} = 75(e^{-(0.22)(1)} - e^{-(2.75)(1)})$$

The whole prefactor has already been collected into the 75, so the only work left is to substitute the time into both exponentials. Each exponent is a reciprocal time multiplied by a time and so is dimensionless.

ALGEBRA

$e^{-0.22} = 0.80252$; $e^{-2.75} = 0.06393$; difference = 0.73859

The absorption exponential has already fallen to about six per cent of its starting value by one hour, which is what makes the curve turn over. The bracket is a pure fraction below one, so the peak must be below the 75.

ALGEBRA

$C_{max} = (75)(0.73859) = 55.394$ mg/L

The intercept of 75 multiplies a dimensionless bracket, so the result keeps the concentration units of the equation. Evaluating at the unrounded time to peak of 0.9983 hour rather than at exactly 1 hour gives the same figure to one decimal place.

ROUND

$C_{max} = 55.4$ mg/L

Report 55.4 milligrams per litre, the value the equation gives at the peak. The tolerance is set narrowly enough to exclude 53.4, which does not follow from the equation at any time near the peak.

> **DOES THE NUMBER MAKE SENSE?**
>
> At t_{max} of about 1 hr the fast term $e^{-2.75}$ is small, 0.06393, and $e^{-0.22}$ is 0.80252, so C_{max} is a little under four fifths of 75 mg/L; it must be below the coefficient 75.

55.4 mg/L is the value the equation gives at the peak. A figure of 53.4 does not follow from it.

Example 2 Worked by the drill

Answer: 6.3 L

WHAT THE STEM GIVES

coefficient	75	in mg/L, set equal to the prefactor and moved to the denominator
k	0.22	the smaller exponent, in hr^{-1}: $k_a - k = 2.53$
k_a	2.75	the larger exponent, in hr^{-1}, in the numerator and in $k_a - k$
D_0	500-mg	in the numerator: $0.87 \times 500 \times 2.75 = 1196.25$
F	87%	as 0.87 in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$75 \text{ mg/L} = \frac{F \cdot k_a \cdot D_0}{V_D(k_a - k)}$$

The leading coefficient of the equation is the entire constant prefactor of the single oral dose equation, so setting it equal to that expression is what makes the volume of distribution recoverable. Every other quantity in the expression is now known.

ALGEBRA

$$V_D = \frac{F \cdot k_a \cdot D_0}{[75 \text{ mg/L} \times (k_a - k)]}$$

The volume of distribution is multiplied by the difference of the rate constants in the denominator, so it is moved up and the 75 is moved down, which is a single cross-multiplication. The rearrangement changes nothing about the relation, only which quantity stands alone.

ALGEBRA

$$V_D = \frac{(0.87)(500 \text{ mg})(2.75 \text{ hr}^{-1})}{[(75 \text{ mg/L})(2.53 \text{ hr}^{-1})]} = \frac{1196.25}{189.75}$$

Milligrams multiplied by reciprocal hours, divided by milligrams per litre multiplied by reciprocal hours, cancels both the milligrams and the reciprocal hours and leaves litres. The 2.53 per hour is the difference between the two rate constants.

ROUND

$$V_D = 6.30 \text{ L}$$

Report 6.3 L; the unrounded division gives 6.3043 litres. Litres is the only unit the arithmetic produces, since milligrams divided by milligrams per litre leaves litres.

> **DOES THE NUMBER MAKE SENSE?**
> The coefficient 75 mg/L equals $F \cdot k_a \cdot D_0$ over $V_D(k_a - k)$. k_a over $(k_a - k)$, 2.75 over 2.53, is a little over 1, so V_D is a little over 0.87×500 over 75 mg/L, about 6 L; 6.30 L fits.

Example 2 Worked by the drill

Answer: 31.11 mg/L

WHAT THE STEM GIVES

coefficient	75	in mg/L, multiplied by the bracket at 4 hr
k	0.22	in hr^{-1}: the exponent $kt = 0.22 \times 4 = 0.88$
k_a	2.75	in hr^{-1}: the exponent $2.75 \times 4 = 11$, a vanishing term

t	4 hours	the time in both exponents

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = 75(e^{-(0.22)(4)} - e^{-(2.75)(4)})$$

A concentration asked for at a stated time needs nothing but substitution of that time into both exponentials, because the constant prefactor has already been collected into the 75. No time to peak is needed, since the question does not concern the turning point.

ALGEBRA

$$e^{-0.88} = 0.41478;\ e^{-11} = 0.0000167$$

Reciprocal hours multiplied by hours gives a dimensionless exponent. By four hours the absorption exponential has fallen to less than two thousandths of one per cent, which is the arithmetic statement that absorption is effectively complete.

ALGEBRA

$$C_p = (75)(0.41478 - 0.0000167) = (75)(0.41477)$$

The difference of the two exponentials is again a pure fraction, so multiplying by the prefactor keeps the units of the prefactor. At this time the absorption term subtracts almost nothing.

ROUND

$$C_p = 31.11 \text{ mg/L}$$

Two decimal places matches the precision her worked concentrations carry in this module. The value is well below the peak of 55.4 mg/L, as it must be four hours after a peak at one hour.

> **DOES THE NUMBER MAKE SENSE?**
>
> By 4 hr the absorption term e^{-11} is gone, so C_p is $75 \times e^{-0.88} = 75 \times 0.41478$, about 31.11 mg/L; 4 hours is a little over one elimination half-life, so a little under half of 75.